

October 25, 2025

Editor-in-Chief
Ecological Informatics
Elsevier

Dear Editor-in-Chief,

I am pleased to submit the manuscript entitled “WS3: An open-source Python framework for integrated simulation and optimization of forest landscape and wood supply systems” for consideration in *Ecological Informatics*. The paper introduces a fully open, solver-agnostic computational workflow that unifies Model I harvest scheduling, libCBM-based carbon accounting, and raster spatial reporting, providing reproducible decision support for forest-sector sustainability planning.

WS3 addresses a persistent informatics gap: professional forest-estate schedulers remain proprietary and opaque, while existing research platforms rarely integrate optimization, carbon models, and geospatial reporting in a scripted, auditable fashion. The manuscript documents the software architecture, mathematical formulation, parity validation against an industrial Woodstock benchmark, and scaling experiments that expose runtime and memory envelopes. It also details the open-source release (MIT-licensed code, Zenodo DOI, reproducible scripts) and highlights how the framework supports cumulative-effects and climate-mitigation analyses—fully aligned with the journal’s emphasis on computational ecology and data-intensive ecosystem management.

The work is original, has not been published previously, and is not under consideration elsewhere. The author has approved the submitted version. Data, code, and reproduction assets are openly archived on Zenodo and GitHub, and highlights, graphical abstract, and other supporting files accompany this submission.

Thank you for considering this submission. I look forward to the opportunity to contribute to *Ecological Informatics*. Please do not hesitate to contact me if additional information is required.

Sincerely,

Gregory Paradis
Faculty of Forestry
University of British Columbia
2424 Main Mall, Vancouver, BC V6T 1Z4, Canada
Email: gregory.paradis@ubc.ca